Frequency-Domain Descriptions for Continuous-Time Linear Time-Invariant Systems

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Key Concepts

1) Frequency response $H(\Omega)$ is the Fourier transform of the impulse response.

    a) The magnitude response is $|H(\Omega)|$.

    b) The phase response is $\arg\{H(\Omega)\}$.

    c) If the input is a complex sinusoid, the output is has its amplitude scaled by $|H(\Omega)|$ and its phase shifted by $\arg\{H(\Omega)\}$.

2) If a system is described by the differential equation

$$\sum_{k=0}^{N} a_k \frac{d^k}{dt^k} y(t) = \sum_{k=0}^{M} b_k \frac{d^k}{dt^k} x(t)$$

then the frequency response is a ratio of polynomials in $j\Omega$

$$H(\Omega) = \frac{\sum_{k=0}^{M} b_k (j\Omega)^k}{\sum_{k=0}^{N} a_k (j\Omega)^k}$$

3) The frequency response of a system described by a differential equation is equal to the transfer function evaluated at $s = j\Omega$.

4) The impulse response of a system described by a differential equa-

tion may be obtained as the inverse Fourier transform of the frequency response.

a) Find the roots s_ℓ of the characteristic equation

$$a_N s^N + a_{N-1} s^{N-1} + \ldots + a_1 s + a_0 = 0 = a_N \prod_{\ell=1}^{N} (s - s_\ell)$$

b) If the roots s_ℓ are distinct and $M < N$, then the partial fraction expansion is

$$H(\Omega) = \frac{A_1}{j\Omega - s_1} + \frac{A_2}{j\Omega - s_2} + \cdots + \frac{A_N}{j\Omega - s_N}$$

c) Solve for the A_i

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d) Take the inverse Fourier transform of each term

$$h(t) = A_1 e^{s_1 t} u(t) + A_2 e^{s_2 t} u(t) + \ldots + A_N e^{s_N t} u(t)$$

e) This requires the roots s_ℓ to have negative real parts, which is also the condition for the system to be stable.

f) If the roots are repeated, a similar, but more complex partial fraction expansion is used.

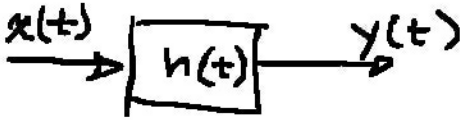
g) If $M \geq N$, then long division must be used prior to partial fraction expansion.

h) If a_k, b_k are real, then the roots s_ℓ and the weights A_ℓ occur in complex conjugate pairs.

Video Screenshots

Time-domain descriptions for linear time-invariant systems

1) Impulse response:



$$y(t) = h(t) * x(t)$$

2) Differential Equation:

$$a_N \frac{d^N}{dt^N} y(t) + a_{N-1} \frac{d^{N-1}}{dt^{N-1}} y(t) + \dots + a_0 y(t) = b_M \frac{d^M}{dt^M} x(t) + \dots + b_0 x(t)$$

Frequency-domain description - Frequency Response $H(\Omega)$

Convolution property

$$y(t) = h(t) * x(t) \xleftrightarrow{FT} Y(\Omega) = H(\Omega) X(\Omega)$$

$$x(t) = A e^{j\Omega_0 t} \xleftrightarrow{FT} X(\Omega) = 2\pi A \delta(\Omega - \Omega_0) \Rightarrow Y(\Omega) = 2\pi A H(\Omega_0) \delta(\Omega - \Omega_0)$$

$$\Rightarrow y(t) = A |H(\Omega_0)| e^{j(\Omega_0 t + \arg\{H(\Omega_0)\})}$$

$|H(\Omega_0)|$: magnitude response , $\arg\{H(\Omega_0)\}$: phase response

Impulse response $h(t) \overset{FT}{\longleftrightarrow} H(\Omega)$ Frequency response

"convolution" "multiplication"

Differential Equations

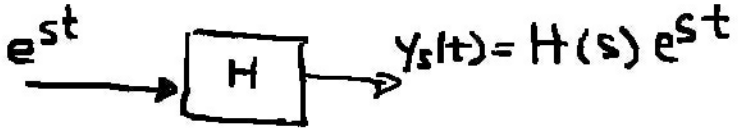
$$\frac{d}{dt} w(t) \overset{FT}{\longleftrightarrow} j\Omega W(\Omega)$$

$$\sum_{k=0}^{N} a_k \frac{d^k}{dt^k} y(t) = \sum_{k=0}^{M} b_k \frac{d^k}{dt^k} x(t) \overset{FT}{\longleftrightarrow} \sum_{k=0}^{N} a_k (j\Omega)^k Y(\Omega) = \sum_{k=0}^{M} b_k (j\Omega)^k X(\Omega)$$

$$H(\Omega) = \frac{\sum_{k=0}^{M} b_k (j\Omega)^k}{\sum_{k=0}^{N} a_k (j\Omega)^k} = \frac{b_M (j\Omega)^M + b_{M-1}(j\Omega)^{M-1} + \cdots + b_0}{a_N (j\Omega)^N + a_{N-1}(j\Omega)^{N-1} + \cdots + a_0}$$

ratio of polynomials in $j\Omega$

Solving differential equations: Steady-state response



$$H(s) = \frac{b_M s^M + b_{M-1} s^{M-1} + \cdots + b_0}{a_N s^N + a_{N-1} s^{N-1} + \cdots + a_1 s + a_0}$$

Transfer function

Frequency Response

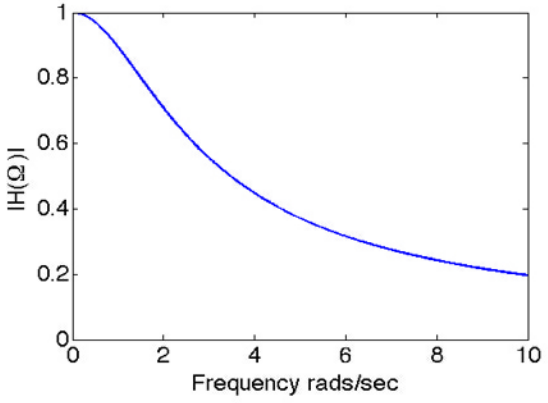
$$H(\Omega) = H(s)\big|_{s=j\Omega}$$

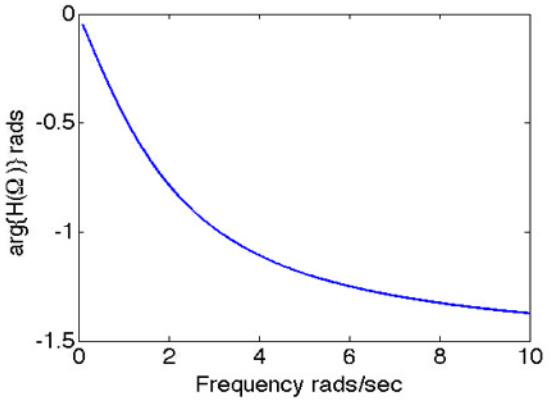
Example:

$$\frac{d}{dt} y(t) + 2\,y(t) = 2\,x(t)$$

$$H(\Omega) = \frac{2}{j\Omega + 2}$$

$N=1,\ M=0,\ b_0=2,\ a_0=2, a_1=1$





Can easily find the impulse response

$$H(\Omega) = \frac{2}{j\Omega + 2} \xleftrightarrow{FT} h(t) = 2e^{-2t}u(t)$$

Finding the impulse response from a differential equation -

$$H(\Omega) \xleftrightarrow{FT} h(t)$$

$H(\Omega)$ is a ratio of polynomials in $j\Omega$: partial fraction expansion

1) Root characteristic equation -

$$a_N s^N + a_{N-1} s^{N-1} + \ldots + a_1 s + a_0 = a_N \prod_{\ell=1}^{N} (s - s_\ell)$$

2) For distinct (non repeated) s_ℓ and $M < N$

$$H(\Omega) = \frac{b_M (j\Omega)^M + b_{M-1}(j\Omega)^{M-1} + \ldots + b_0}{a_N (j\Omega - s_1)(j\Omega - s_2)\cdots(j\Omega - s_N)} = \frac{A_1}{j\Omega - s_1} + \frac{A_2}{j\Omega - s_2} + \cdots + \frac{A_N}{j\Omega - s_N}$$

3) Invert each term

$$h(t) = A_1 e^{s_1 t} u(t) + A_2 e^{s_2 t} u(t) + \cdots + A_N e^{s_N t} u(t)$$

Example: $\frac{d^2}{dt^2}y(t) + 5\frac{d}{dt}y(t) + 6y(t) = 5\frac{d}{dt}x(t) + 12x(t)$

$$H(\Omega) = \frac{5j\Omega + 12}{(j\Omega)^2 + 5j\Omega + 6}$$

$s^2 + 5s + 6 = (s+3)(s+2)$ $\quad s_1 = -3$, $s_2 = -2$

Expand $\frac{5j\Omega + 12}{(j\Omega)^2 + 5j\Omega + 6} = \frac{A_1}{j\Omega + 3} + \frac{A_2}{j\Omega + 2} \Rightarrow \begin{matrix} A_1 + A_2 = 5 \\ 2A_1 + 3A_2 = 12 \end{matrix}$ $\quad A_1 = 3, A_2 = 2$

$$h(t) = 3e^{-3t}u(t) + 2e^{-2t}u(t)$$

Comments –

1) Repeat roots follows same approach – messier algebra
2) If $M \geq N$, use long division to obtain proper rational function
3) For real a_k, b_k: complex roots occur in conjugate pairs
   expansion coefficients A_k must be conjugates